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1 PURPOSE AND GUIDANCE

This document is made to outline the discussions with **Attalon** on the CRIMs (gaps) carried out related to IFS® functionality. In addition, this document lays out the processes **Attalon** practice currently and possible changes to the workflows and possible solutions under the IFS®.

This document specifies:

- Attalon business requirements
- How screens, views, fields, and tables shall look and operate.
- How business logic and functions shall operate and/or calculate
- How the application shall be customized to solve the business requirements.

The **Solution Description** defines **what** will be delivered. The solution is matched against the requirements described in the Requirements section.

The **Delimitations** describe what is explicitly **not covered** by this CRIM. Together with the Solution description, these sections delimit the extent of the functional addition agreed for implementation.

2 OPEN ISSUES/DECISIONS REQUIRED

This specification cannot be completed until the following issues/decisions have been resolved:

Issue/Decision	Responsible

3 BUSINESS REQUIREMENTS DETAILS

3.1 BUSINESS REQUIREMENTS

- Attalon requires that IFS Freight Interface functionality, APIs, Basic Data Tables and interface to the larger IFS ERP applications be leveraged as much as possible for this specification. The primary development effort is leveraging the "Freight Provider" functionality in IFS for purpose of interfacing with UPS for quoting freight cost and price.
 - Existing Freight Interface in IFS includes:
 - Shipment Management / Freight Integration
 - Basic Data
 - Freight Provider (i.e., use provider Pacejet to understand how to setup provider UPS)
 - In separate specification (I-007) replace with UPS
 - Shipment Services
 - Examples are overnight, next day, etc.
 - Freight Carriers
 - Examples are: UPS, UPS, UPS Over Night, ect.
 - Freight Delivery Terms
 - Examples are pre-paid and add (PPAD)
 - Freight Transactions
 - All Financial transactions for the freight interface.

Specification

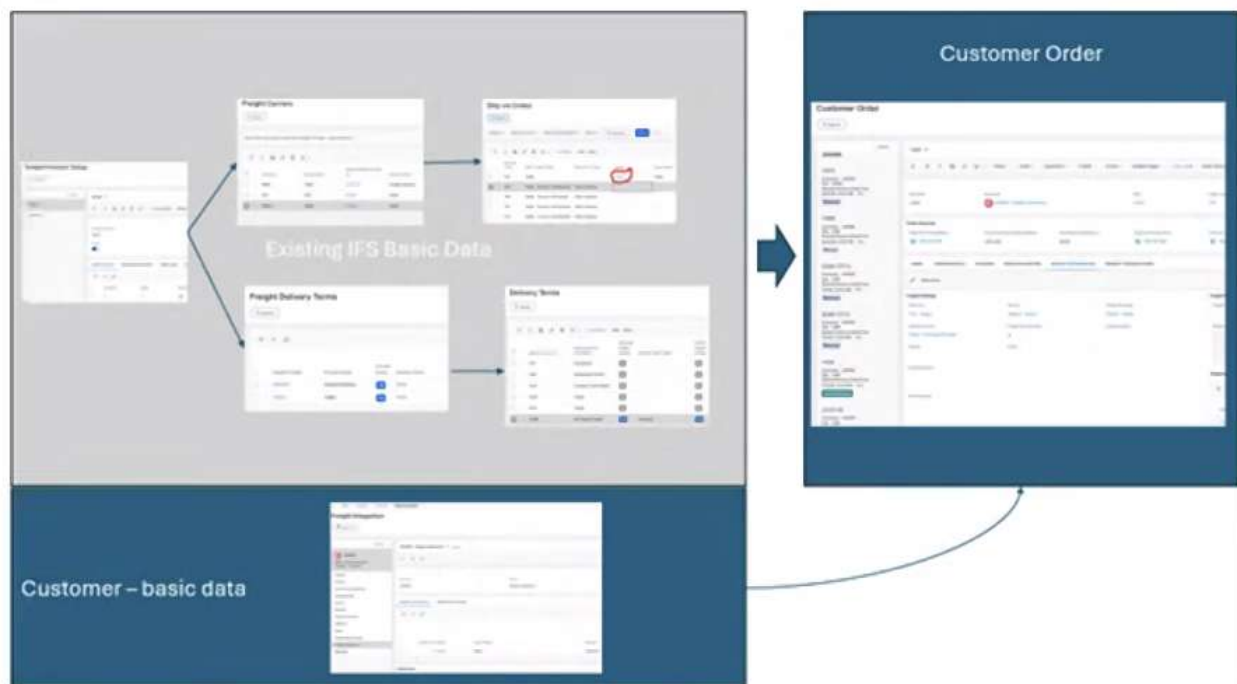
- May include calls / return payloads from UPS or any provider
- Freight Log
 - All calls / return payloads from UPS or any provider
- Attalon requires that at Shipment (Shipment Header Screen), the shipping department will be able to automate a call to either UPS or UPS and receive the return values for cost and price for a specific shipment and service.
- Attalon requires that the shipment cost and price be able to be marked up (%)
- Attalon requires that the 'selected' shipment service (UPS) be able to be selected and then inserted into the original customer order charge line and linked to the customer order line found on the first shipment line of the shipment.
 - The COL charges be updated or inserted with the cost and markup price.
- Attalon requires that an integration to UPS (and FedEx) be able to call UPS (and FedEx) open interfaces (RESTful API) and retrieve information regarding shipment.
 - Interface (POST) will send to UPS (and FedEx)
 - Deliver From Address
 - Deliver To Address
 - Service Type (i.e. overnight, same-day, ground, etc.)
 - Handling Unit/s for Shipment (Pallets, Boxes, etc.)
 - Type Handling Unit/s (Pallet, Box)
 - Size Handling Unit/s (L, W, H)
 - Weight of Handling Unit/s
 - Shipment Date
 - Special services: overnight, ...
 - Interface will receive an initial estimate for booking to the Shipment Freight Transactions Line
 - Shipment Reviews Estimate and then once estimate is selected, it will book the UPS shipment assigning the correct charge to the correct charge line associated with the correct customer order line.
- Attalon requires that returned UPS 'estimate' be able to be selected and booked.
 - The Customer Order Charge Line is created which is linked to the First Customer Order Line found on the shipment.
 - The charge cost is the cost of freight in return payload from UPS cost
 - The charge price will be the charge cost x markup found on the Freight Delivery Terms.
 - The sales charge used on the CO charge line will be found in the Freight Delivery Terms.
 - The Shipment Header is updated
 - Planned Ship Date is updated with Pickup date, time
 - PRO No is updated with Tracking Number (master or first tracking No)
- Attalon requires that a background job run once per day to transmit to the customer that the shipment has occurred. It will only transmit those shipments, shipped (i.e., delivered) that day to the customer.

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Specification

4 SOLUTION DESCRIPTION

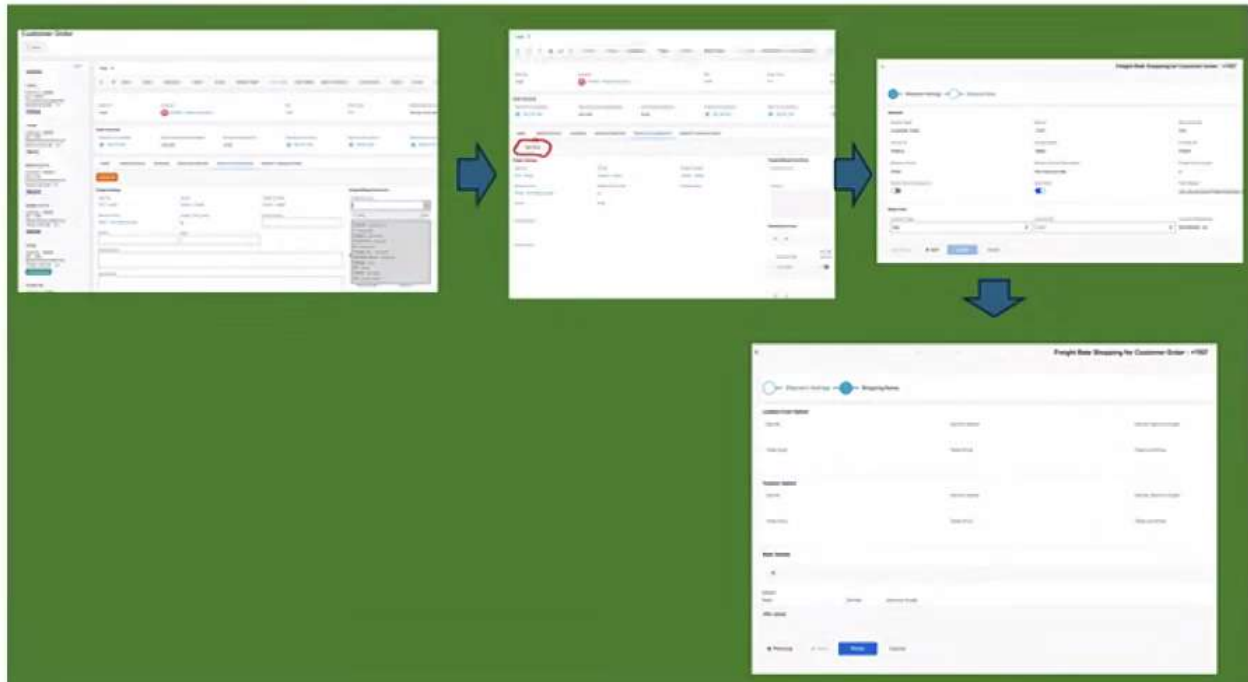
4.1 IFS FREIGHT INTERFACE BASIC DATA SETUP



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Specification

4.2 PROCESS FLOW FOR INTERFACE



4.3 SETUP FREIGHT PROVIDER SCREEN

Using the Freight Provider Screen, create a new Freight Provider "UPS". Determine the correct config values, provider locations and terms code (for freight delivery terms) as well as mapping messages:

- Config Values:
 - Values used to connect to the correct UPS APIs, username, password, passcode, secrete, mappings to the end-projections or end APIs at UPS.
 - Try to find a similar setup to UPS and use these as the basic mapping data.
- Terms Code
 - These are used to connect the delivery terms used by UPS to the Freight Delivery Terms used in IFS. The mapping is one to one.
- Provider Locations
 - These are the site locations in IFS mapped to the account/site/warehouse locations at UPS (or freight provider).
- Message Mapping
 - For Calls and Return Payloads, message mapping allows for translation between UPS and IFS.

Attalon

🏠 > Shipment Management > Freight Integration > Basic Data > Freight Provider Setup

Freight Provider Setup

🔍 Search

FEDEX

PACEJET

FEDEX

Freight Provider ID

FEDEX

Active

🔴

Name*

FedEx

Default Provider

Yes

CONFIG VALUES PROVIDER LOCATIONS TERMS CODE MESSAGE MAPPING

☰ + ✎

✓	Config ID	Value	Description
☐			

Set as Default Validate Interface Parameters

4.4 PROCESS FLOW BUSINESS LOGIC

- Initial Call to UPS
 - Organize Payload based on L, W, H of boxes (or pallets)
 - Total Qty of boxes (or pallets) by L, W, H
 - Total Weight of boxes (or pallets) by L, W, H
 - Ship from address (on shipment header), ship to address (on shipment header), and service
 - Shipment Date
 - Call UPS → Send Payload
 - Receive Return Payload
 - Insert Return Payload into the Shipment / Tab "Freight Transactions" Screen.
 - If return is an array of potential pickup dates, times, and/or services then insert into multiple lines in table.
- Select and Book
 - (Note, that this may require a secondary confirmation/booking call to UPS)
 - Using delivery terms on the shipment, lookup freight delivery terms. Using the freight delivery terms lookup the sales charge linked to the freight delivery terms.
 - Using the sales charge and the source ref 1, 2, 3, 4 of the first shipment line, insert the sales charge on the customer order linked to the customer order line.
 - Using the cost in the return payload, calculate the price by using markup x price. (i.e. $\$100 \times 1.5 = \150)
 - Insert or update the charge line cost and price
 - Update the shipment header / General / Reference Documents and Misc. Info/ Field PRO No with the tracking number.
 - Update the shipment header / Planned Ship Date Time with the estimated pickup date and time in return payload.

4.5 UPS DEV

5 UPS Open APIs for ERP Integration

UPS offers a suite of **open APIs** and integration options that can connect directly to your **Enterprise Resource Planning (ERP)** system, enabling seamless shipping, tracking, and supply chain data exchange.

5.1 Key Integration Capabilities

RESTful APIs: UPS provides standardized REST APIs for real-time shipping, rate, transit, and tracking data [UPS](#).

Forwarding APIs: For real-time freight forwarding automation, allowing direct integration with ERP/TMS/WMS systems [UPS](#).

EDI Integration: For high-volume, standardized data exchange (e.g., shipment documents, transactions) between UPS and ERP systems [UPS](#).

Hybrid Integration: Combines real-time API connections with batch processing for flexibility [UPS](#).

UPS Supply Chain Symphony™ APIs: For visibility and orchestration across systems, including ERP, TMS, WMS, and analytics platforms [UPS](#).

5.2 Integration Benefits

Automated workflows: Shipments, quotes, and labels can be created and processed directly from ERP.

Real-time visibility: Track shipments and monitor supply chain status from within ERP.

High-volume support: EDI and API solutions handle large transaction volumes without system disruption [UPS](#).

Standardized formats: EDI ensures compatibility with existing ERP data models; APIs use JSON/XML for flexible integration.

5.3 How to Connect to ERP

Access the UPS Developer Portal to browse the API catalog, request credentials, and review documentation [UPS Developer Portal](#)¹.

Choose the right integration type:

API-first: For real-time, lightweight integrations.

EDI: For structured, high-volume transactions.

Hybrid: For mixed real-time and batch needs.

Set up credentials (OAuth, client ID, client secret, shipper number) via the Developer Portal [Postman](#).

Map data fields between your ERP system and UPS APIs/EDI schemas.

Test and monitor using the API Status Dashboard to ensure uptime and performance [UPS](#).

5.4 Example Use Cases

E-commerce ERP: Integrate UPS APIs to provide customers with real-time shipping rates and tracking.

Manufacturing ERP: Automate inbound/outbound freight booking and document exchange via EDI.

Logistics ERP: Combine API-based shipment creation with EDI for consolidated reporting.

For full details, start with the [UPS Developer Portal](#) and the [Developer Resource Center](#) to select the APIs and integration methods that best fit your ERP architecture.

5.5 EXAMPLE OF PAYLOAD TO UPS

Message Sent --> IFS to UPS

Example UPS Freight Rate Request Payload (JSON)

Json

```
{
  "FreightRateRequest": {
    "Request": {
      "RequestOption": "Rate",
      "TransactionReference": {
        "CustomerContext": "Freight rate quote request"
      }
    },
    "ShipFrom": {
      "Name": "Sender Company",
      "Address": {
        "AddressLine": ["123 Origin St"],
        "City": "Atlanta",
        "StateProvinceCode": "GA",
        "PostalCode": "30301",
        "CountryCode": "US"
      }
    },
    "ShipTo": {
      "Name": "Receiver Company",
      "Address": {
        "AddressLine": ["456 Destination Ave"],
        "City": "Chicago",
        "StateProvinceCode": "IL",

```

Specification

```
"PostalCode": "60601",
"CountryCode": "US"
},
"PaymentInformation": {
  "Payer": {
    "Name": "Payer Company",
    "Address": {
      "AddressLine": ["789 Billing Rd"],
      "City": "Dallas",
      "StateProvinceCode": "TX",
      "PostalCode": "75201",
      "CountryCode": "US"
    }
  },
  "ShipmentBillingOption": {
    "Code": "10",
    "Description": "Prepaid"
  }
},
"Service": {
  "Code": "308",
  "Description": "UPS Freight LTL"
},
"HandlingUnitOne": {
  "Quantity": "1",
  "Type": {
    "Code": "PLT",
    "Description": "Pallet"
  }
},
"Commodity": [
  {
    "Description": "Machinery Parts",
    "Weight": {
      "UnitOfMeasurement": {
        "Code": "LBS"
      },
      "Value": "500"
    },
    "NumberOfPieces": "1",
    "PackagingType": {
      "Code": "PLT"
    },
    "FreightClass": "70"
  }
]
```

Specification

```
}  
]  
}  
}
```

Key Notes:

- **Service Code:** "308" is an example for UPS Freight LTL; use the correct code from UPS documentation.
- **FreightClass:** Must match NMFC classification for your commodity.
- **PaymentInformation:** Adjust "Code" for billing type (e.g., prepaid, collect, third party).
- **HandlingUnitOne:** Defines the type and quantity of handling units (pallets, skids, crates, etc.).

?

6 SCREEN SHOT EXAMPLES

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Specification

Home > Shipping > Shipment Delivery > Shipment > 14798

Shipment

Search Advanced Search with Time Zone

Status Shipment ID Notes Receiver ID Sender Address ID More Favorites Search Saved Searches

Sort by

Receiver ID: 100299
Receiver Description: AutoZone

194838
Cancelled
Receiver ID: 104501
Receiver Description: VusionGroup, Inc.

4682
Cancelled
Receiver ID: LH-INTERNAL-CUST
Receiver Description: Loves Park Internal Customer

3648
Cancelled
Receiver ID: LH-INTERNAL-CUST
Receiver Description: Loves Park Internal Customer

2857
Cancelled
Receiver ID: 100299
Receiver Description: AutoZone

14798
Cancelled
Receiver ID: ROC-INTERNAL-CUST
Receiver Description: Rockford Internal Customer

Sender

Sender Type	Sender ID	Sender Description
Site	CHS	Del Plains

Receiver

Receiver Type	Receiver ID	Receiver Description
Customer	ROC-INTERNAL-CUST	Rockford Internal Customer

Settings and Indicators

SHIPMENT LINES AVAILABLE LINES ADDRESS GENERAL FREIGHT INTEGRATION FREIGHT TRANSACTIONS

Rate Shop

Freight Settings

Ship Via	Carrier	Freight Provider
217 CH ROBBINS GROUND	CH ROBBINS - CH Robbins	NOT RATED - Unrated Carrier

Delivery Terms: DAPN - Deliver Place/Address... Prepaid

Phone: Email:

Shipping Name:

Internal Note:

Freight Billing/Third Party

Freight Account:

Address:

Open Map

Shipping Services

Service Code Include Service

(No data)

Home > Shipping > Shipment Delivery > Shipment > 14798

Shipment

Search Advanced Search with Time Zone

Status Shipment ID Notes Receiver ID Sender Address ID More Favorites Search Saved Searches

Sort by

Receiver ID: 100299
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Cancelled
Receiver ID: 100299
Receiver Description: AutoZone

14798
Cancelled
Receiver ID: ROC-INTERNAL-CUST
Receiver Description: Rockford Internal Customer

Shipment ID: 14798
Site: CHS
Source Ref Type: Customer Order
Next Step in Shipment Flow

Shipment Type: N/A
Created: 2/10/2026
Planned Ship Date/Time: 2/10/26, 12:00 AM CST
Planned Delivery Date/Time

Notes: Dispatch Advice history

Sender

Sender Type	Sender ID	Sender Description
Site	CHS	Del Plains

Receiver

Receiver Type	Receiver ID	Receiver Description
Customer	ROC-INTERNAL-CUST	Rockford Internal Customer

Settings and Indicators

SHIPMENT LINES AVAILABLE LINES ADDRESS GENERAL FREIGHT INTEGRATION FREIGHT TRANSACTIONS

Freight Transactions

Send to NOT RATED

State	Error	Freight Message Type	Freight Provider ID	Delivery Link	Create Date	Total Weight	Total Price	Total Cost	Total Unit Price	Currency Code
(No data)										

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Specification

Shipment Management > Freight Integration > Basic Data > Freight Delivery Terms

Freight Delivery Terms

Search

	Freight Provider	Provider Name	Provider Active	Delivery Terms	Delivery Terms Description	Default Freight Payer	Freight Terms Code	Freight Terms Code Description	Add Estimated Freight Cost	Add Estimated Freight Price	Invoice Estimated Freight	Add Actual Freight Cost	Add Actual Freight Price	Invoice Actual Freight
	FEDX	FedEx Web Services	Yes	DDPN	DeliverOnPaidDest...		Prepaid	Prepaid	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	FCAR	FreeCarrierOriginAdd...		Prepaid	Prepaid	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	DDPDP	DeliverPaidDesUnb...		Third Party	Third Party	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	DDPCD	DeliverPaidDestAdd...		Freight Collect	Freight Collect	Yes	Yes	Yes	Yes	Yes	Yes
	FREIGHTWISE	Freightwise LLC	Yes	DDPN	DeliverOnPaidDest...		Prepaid	Prepaid	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	DDPDP	DeliverOnPaidDest...		Third Party	Third Party	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	DDPCD	DeliverOnPaidDest...		Freight Collect	Freight Collect	Yes	Yes	Yes	Yes	Yes	Yes
	FREIGHTWISE	Freightwise LLC	Yes	FCAR	FreeCarrierOriginAdd...		Prepaid	Prepaid	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	DDPDP	DeliverOnPaidDest...		Third Party	Third Party	Yes	Yes	Yes	Yes	Yes	Yes
	FEDX	FedEx Web Services	Yes	FCAR	FreeCarrierOriginAdd...		Third Party	Third Party	Yes	Yes	Yes	Yes	Yes	Yes

Shipment Management > Freight Integration > Basic Data > Freight Carriers

Freight Carriers

Search

Carrier ID must match value from Freight Provider - case sensitive.

	Carrier ID	Carrier Name	Invoicing Supplier ID	Supplier Name	Default Freight Provider ID	Provider Name	SCAC Code	Carrier
	MATSON NAVIGA...	Matson Navigation			NOT RATED	Unrated Carrier	MATN	Yes
	RST TRANSPORT	RST Transport			NOT RATED	Unrated Carrier	RST	Yes
	SATURN TRUCKING	Saturn Trucking			NOT RATED	Unrated Carrier	SATT	Yes
	TRANSPACE	Transpace			NOT RATED	Unrated Carrier	TRSP	Yes
	DT EXPRESS	DT Express			NOT RATED	Unrated Carrier		Yes
	PANTHOR EXPED...	Panthor Expedited Services			NOT RATED	Unrated Carrier	PTHT	Yes
	EMERGENTECH INC	Emergentech, Inc.			NOT RATED	Unrated Carrier	EMOV	Yes
	BLAN TRANSPOR...	Blan Transportation Management Systems			NOT RATED	Unrated Carrier	BLAN	Yes

Shipment Management > Freight Integration > Basic Data > Shipment Services

Shipment Services

Search

	Service Code	Service Name	Service Type	Service Usage	Provider Tag Name	Provider Tag Type	Accepted Tag Values	Display Sequence	Notes	Validity
	SATURDAY	Saturday Delivery	Shipment Option	Optional	Saturday	Boolean	0,1	1	/1/1/01/05/25	Active

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Specification

Document Management

Freight Integration

Basic Data

Freight Provider Setup

Freight Provider Setup

Search

Sort By

FEDEX

Freight Provider ID

FEDEX

Active

Name

FedEx Web Services

Ship Contact Role

Ship Contact Role Name

Default Provider

Yes

COMING VALUES

PROVIDER LOCATIONS

TERMS CODE

MESSAGE MAPPING

Config ID

Value

Description

ACCOUNT_NUMBER

108923660

MAX_CARTON_CAPACITY

10

MAX_WEIGHT_CAPACITY

200

Document Management

Freight Integration

Basic Data

Freight Provider Setup

Freight Provider Setup

Search

Sort By

FEDEX

Freight Provider ID

FEDEX

Active

Name

FedEx Web Services

Ship Contact Role

Ship Contact Role Name

Default Provider

Yes

COMING VALUES

PROVIDER LOCATIONS

TERMS CODE

MESSAGE MAPPING

Site

Sender Type

Sender ID

Sender Description

Freight Site Code

Freight Facility

Note

(No data)